

COMP 2121 – Quiz 4

Time: 20 min.

Possible marks: 5

Name:

Student #:

1. (2 marks) Given

$$\sum_{k=44}^{100} a_{k+1} = 104 \quad \text{and} \quad \sum_{i=47}^{54} a_{i-2} + \sum_{j=53}^{102} (a_{j-1} + 2) = 126$$

find a_{52} .

$$\sum_{k=44}^{100} a_{k+1} = 104$$

$$\sum_{k=45}^{101} a_k = 104$$

$$\sum_{i=47}^{54} a_{i-2} + \sum_{j=53}^{102} (a_{j-1} + 2) = 126$$

$$\sum_{i=45}^{52} a_i + \sum_{j=53}^{102} a_{j-1} + \sum_{j=53}^{102} 2 = 126$$

$$\sum_{i=45}^{52} a_i + \sum_{j=52}^{101} a_j + 2(102 - 53 + 1) = 126$$

$$a_{52} + \sum_{i=45}^{51} a_i + \sum_{j=52}^{101} a_j + 2(50) = 126$$

$$a_{52} + \sum_{i=45}^{101} a_i = 26$$

Substitute in

$$a_{52} + 104 = 26$$

$$a_{52} = -78$$

2. (3 marks) Use formula $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ to find the value of counter after the following programming segment is executed

```

counter = 200;
for (i = 6; i <= n+3; i++){
    counter = counter + 4i
    for (j = i; j <= n+1; j++){
        counter = counter + 6
    }
}
//assume n>5

```

Simplify your answer and far as possible.

$$\begin{aligned}
 \text{Counter} &= 200 + \sum_{i=6}^{n+3} \left(4i + \sum_{j=i}^{n+1} 6 \right) \\
 &= 200 + \sum_{i=6}^{n+3} (4i + 6(n+1-i+1)) \\
 &= 200 + \sum_{i=6}^{n+3} (-2i + 6n + 12) \\
 &= 200 - 2 \sum_{i=6}^{n+3} i + \sum_{i=6}^{n+3} (6n + 12) \\
 &= 200 - 2 \left(\sum_{i=1}^{n+3} i - \sum_{i=1}^5 i \right) + (6n + 12)(n + 3 - 6 + 1) \\
 &= 200 - 2 \left[\frac{(n+3)(n+4)}{2} - \frac{5(6)}{2} \right] + (6n + 12)(n - 2) \\
 &= 200 - (n+3)(n+4) + 30 + 6n^2 - 24 \\
 &= 200 - n^2 - 7n - 12 + 30 + 6n^2 - 24 \\
 &= 5n^2 - 7n + 194
 \end{aligned}$$
